Rezaul Islam

Dhaka-1000, Bangladesh | rezaul90@iwfm.buet.ac.bd

Academic Credentials

Bangladesh University of Engineering and Technology (BUET) — Ongoing
Master of Science (M.Sc.) in Civil & Structural Engineering — Dhaka, Bangladesh

Bangladesh University of Engineering and Technology (BUET) — 2023
Bachelor of Science (B.Sc.) in Civil Engineering — Dhaka, Bangladesh
Major: Structural Engineering

Research Interests

Coastal Engineering and Wave–Structure Interaction, Wave Mechanics and Coastal Hydraulics, Computational Fluid Dynamics (CFD) and Numerical Modeling, Hydraulic Structures and Coastal Protection Infrastructure, Hydrological Modeling, Remote Sensing and GIS Applications, Structural Mechanics and Finite Element Analysis

Professional Experience

Lecturer — December 2023 – Present
Institute of Water and Flood Management (IWFM), BUET — Dhaka, Bangladesh

Research Experience

M.Sc. Thesis — Ongoing
Supervisor: Dr. Raquib Ahsan, Professor, Department of Civil Engineering, BUET
Co-Supervisor: Dr. Mohammad Asad Hussain, Professor, Institute of Water and Flood Management (IWFM), BUET
Topic: Numerical Study of Seawall Geometries for Enhanced Structural Performance under Hydrodynamic Wave Action

Undergraduate Thesis — 2023
Supervisor: Dr. Tahsin Reza Hossain, Department of Civil Engineering, BUET
Topic: Performance Evaluation of RC Moment Frames Using Nonlinear Pushover Analysis

Research Projects

Co-Investigator — November 2023 – June 2027
Implementation of Geospatial Technique in Flood and Drought Monitoring for the Safety of Nuclear Installations
Funded by: International Atomic Energy Agency (IAEA)
Lead: Bangladesh Atomic Energy Commission (BAEC); Affiliate: IWFM, BUET

Co-Investigator — November 2024 – April 2026
Anticipatory Action for Health in a Changing Climate
Funded by: Tufts University, USA and The Rockefeller Foundation, USA

Co-Investigator — April 2025 – December 2025
Application and Updating of the CCM for Cyclone Impact Prediction and the DFRM for the Risk-based Riverine Flood Warning in Bangladesh
Funded by: Eco-Social Development Organization (ESDO)

Co-Investigator — October 2022 – June 2025
Enhancing Climate Resilience of Coastal Areas of Bangladesh with Nature Based Solutions
Funded by: Global Center on Adaptation (GCA); Lead: IWFM, BUET

Advisory Services

Team Member — October 2023 – September 2025
Hydro-Morphology Study of Bridges under Dhaka and Mymensingh Divisions
Client: Local Government Engineering Department (LGED)

Publications

Conference

[1] Tushar, M. N., **Islam, R.**, Muktadir, H. M., Shampa, M., Rahman, M. M., Islam, A. K. M. S., & Hussain, M. A. (2025). Assessing the replicability of bandal-like structures for sediment management in the Kuakata region of the coastal zone of Bangladesh. Proceedings of the 10th International Conference on Water and Flood Management (ICWFM 2025), Dhaka, Bangladesh.

[2] **Islam, R.**, Tushar, M. N., Dey, S., Uddin, S. I., Shampa, M., Chowdhury, A. I. A., Murshed, S. B., Haque, S., & Salehin, M. (2025). Predicting Ambient Air Temperature of Bangladesh Using Machine Learning and Remote Sensing. AGU Fall Meeting Abstracts, 2025, A31G-VR8865.

[3] Billah, M. M., Sakib, M. R., & **Islam, R.** (2025). Multi-layer perceptron Markov chain-based artificial neural network for modeling future land-specific carbon emission patterns and its effects on land surface temperature in Rajshahi City. AGU Fall Meeting Abstracts, 2025, GC41N-VR8883.

Technical Skills

Structural Analysis and Design: ANSYS Mechanical (Static Structural, Transient Structural), ETABS, SAP2000

Computational Fluid Dynamics: ANSYS Fluent, CFD-Post

GIS and Remote Sensing: ArcGIS, raster analysis, GIS-based spatial analysis

Hydrological Modeling: HEC-HMS

Programming and Scientific Computing: Python, C++, MATLAB, Mathematica

Machine Learning and Data Analysis: Machine learning, Python-based data analysis, Microsoft Excel

Climate and Geospatial Data Processing: NetCDF processing, climate data analysis, geospatial data processing

Involvements

Co-Instructor April 2025 Semester
Postgraduate seminar course, IWFM, BUET

Assistant Organizing Secretary 2025
International Conference on Water and Flood Management (ICWFM) 2025, IWFM, BUET

Honors and Awards

University Merit Scholarship, BUET
Awarded for academic excellence in undergraduate studies

Dean's List Award, Department of Civil Engineering, BUET
Awarded for academic excellence across all four undergraduate levels (I–IV)